

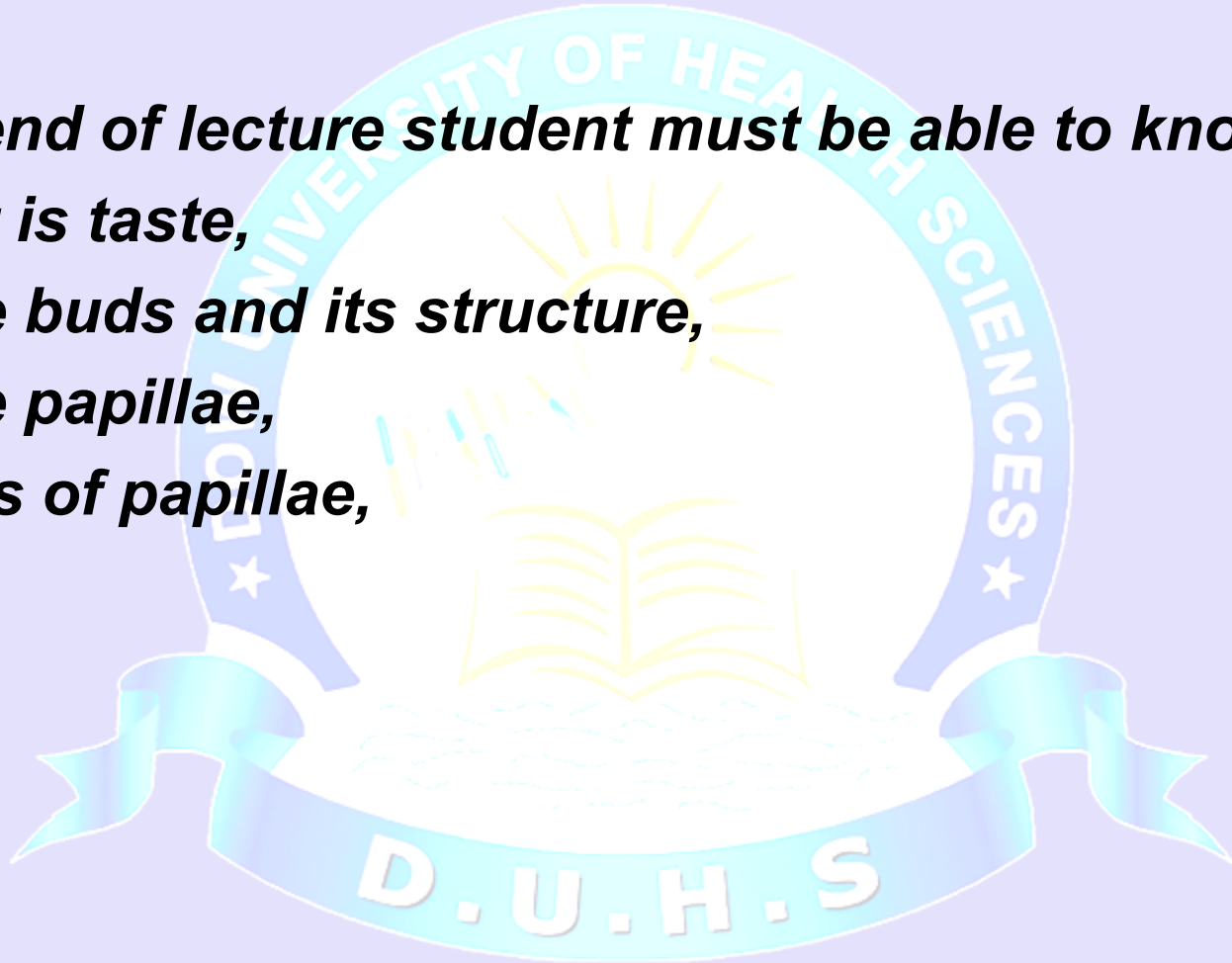
HISTOLOGY OF TONGUE



“LEARNING OBJECTIVE”

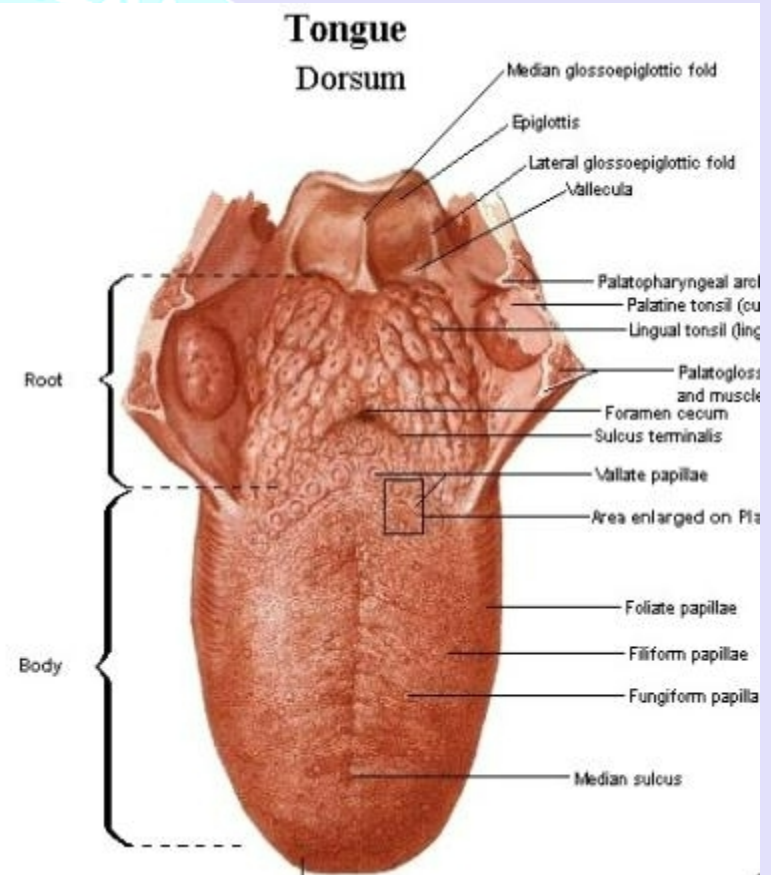
At the end of lecture student must be able to know,

- ***What is taste,***
- ***Taste buds and its structure,***
- ***Taste papillae,***
- ***Types of papillae,***



TONGUE

- It is a mass of **skeletal muscle** covered on both side with **mucosa**.
- **Muscles** fibers in the tongue run in three directions, longitudinal, transverse and vertical.
- The **mucosa** in the anterior part is rough due to the presence of projections called **papilla**.
- In the posterior part the mucosa contains **lymphoid follicle**.



TONGUE

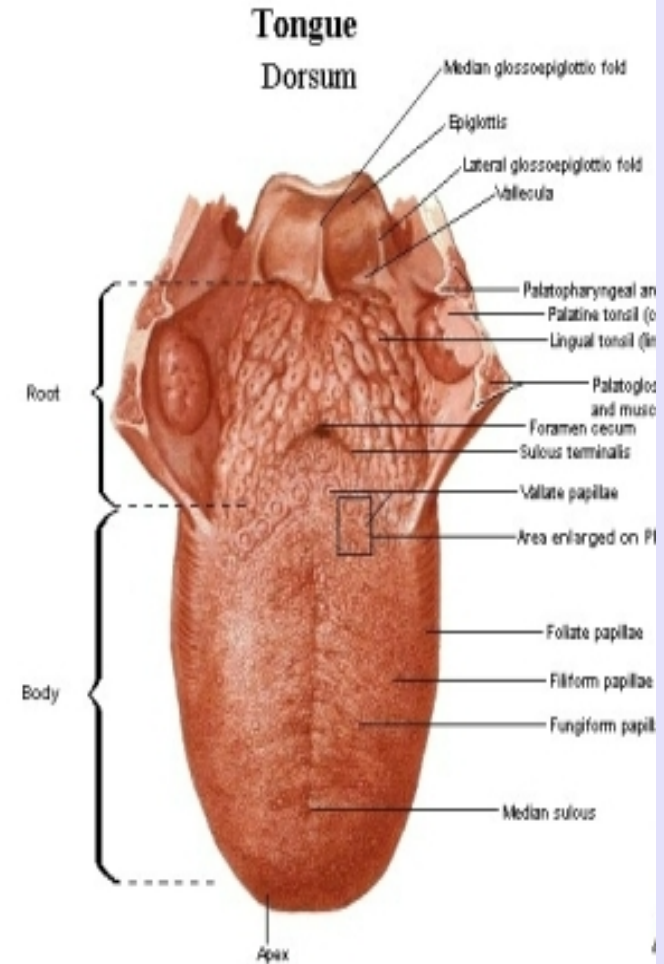
◎ *The dorsal surface of the tongue is divided by a V-shaped sulcus terminalis into:*

*1- An **oral part**, the anterior two-thirds.*

*2- A **pharyngeal part**, the posterior one-third.*

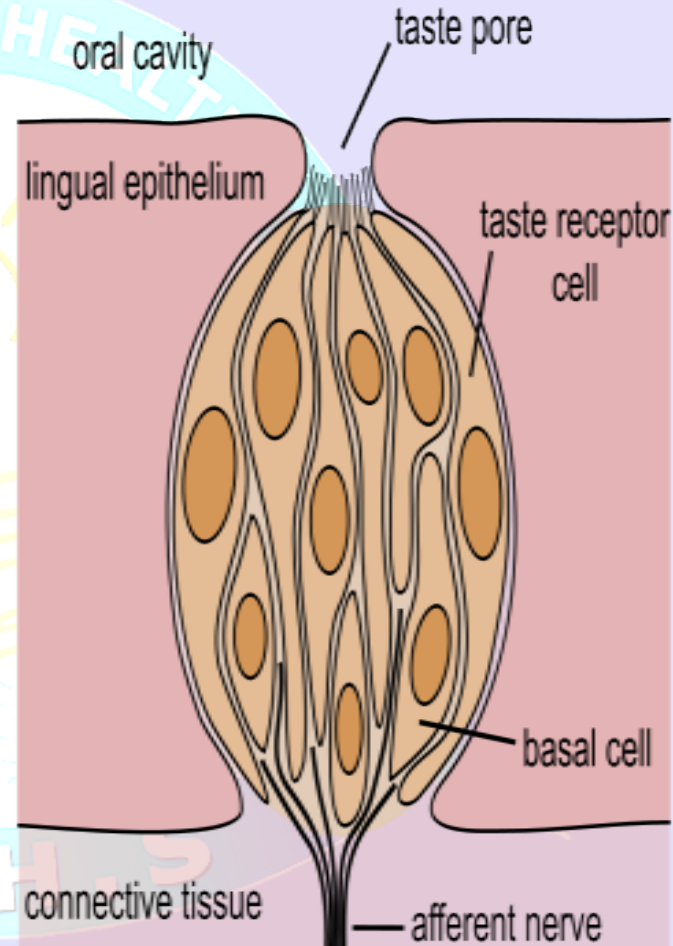
◎ *The dorsal surface of the oral part has the lingual papillae.*

◎ *The lingual papillae consist of a connective tissue core covered with stratified squamous epithelium.*



“TASTE BUDS”

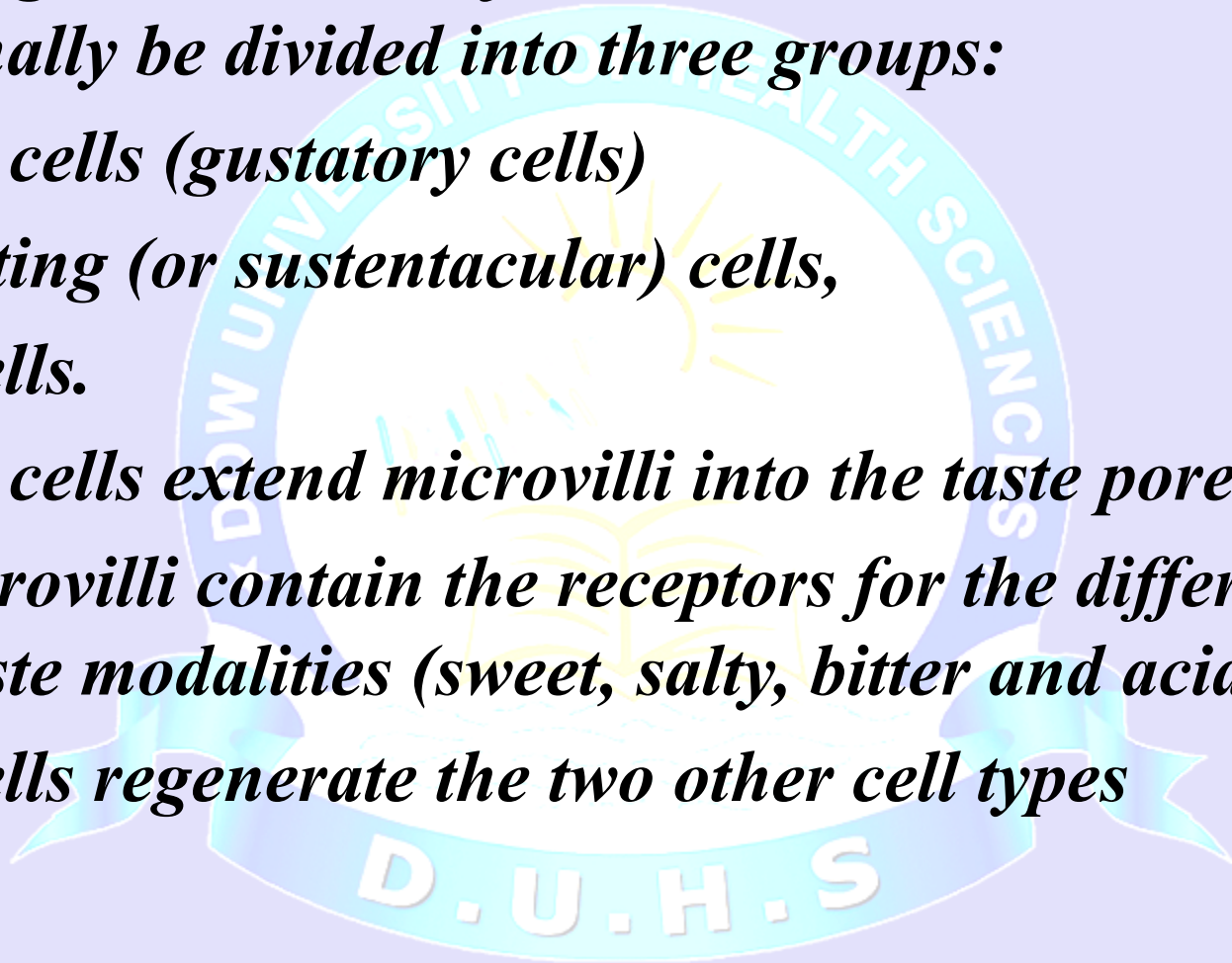
- ***Taste buds contain the receptors for taste.***
- ***They are located around the small structures on the upper surface of the tongue, soft palate, upper esophagus and epiglottis, which are called papillae.***
- ***These structures are involved in detecting the five (known) elements of taste perception: salty, sour, bitter, sweet, and savory (or umami).***



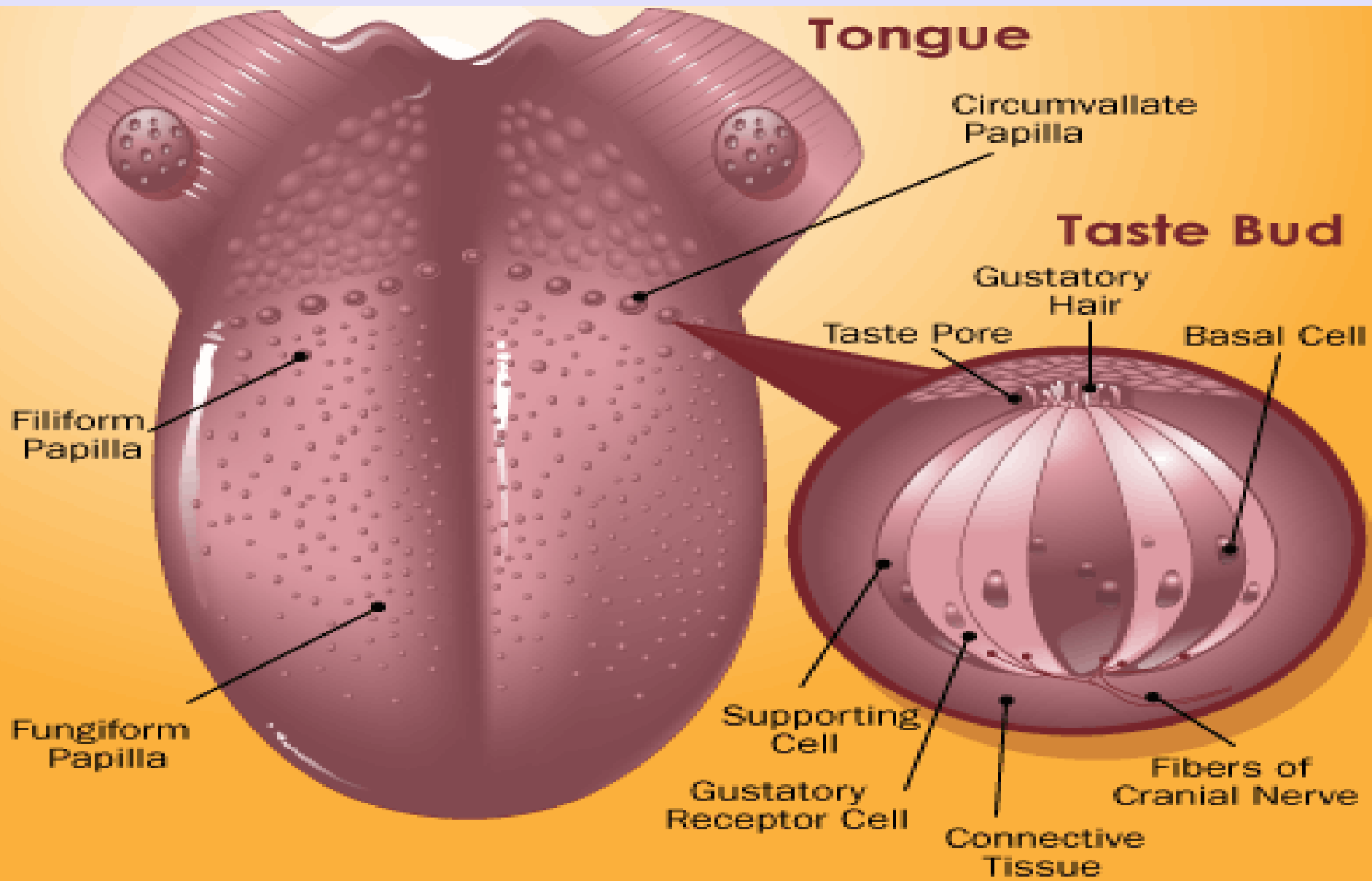
“TASTE BUDS”

- ◎ ***Via small openings in the tongue epithelium, called taste pores, parts of the food dissolved in saliva come into contact with taste receptors.***
- ◎ ***These are located on top of the taste receptor cells that constitute the taste buds.***
- ◎ ***The taste receptor cells send information detected by clusters of various receptors and ion channels to the gustatory areas of the brain via the seventh, ninth and tenth cranial nerves.***
- ◎ ***On average, the human tongue has 2,000–8,000 taste buds.***

- *The elongated cells that form the taste bud can functionally be divided into three groups:*
 1. *Sensory cells (gustatory cells)*
 2. *Supporting (or sustentacular) cells,*
 3. *Basal cells.*
- *Sensory cells extend microvilli into the taste pore.*
- *The microvilli contain the receptors for the different basic taste modalities (sweet, salty, bitter and acid).*
- *Basal cells regenerate the two other cell types*



“TASTE BUDS”



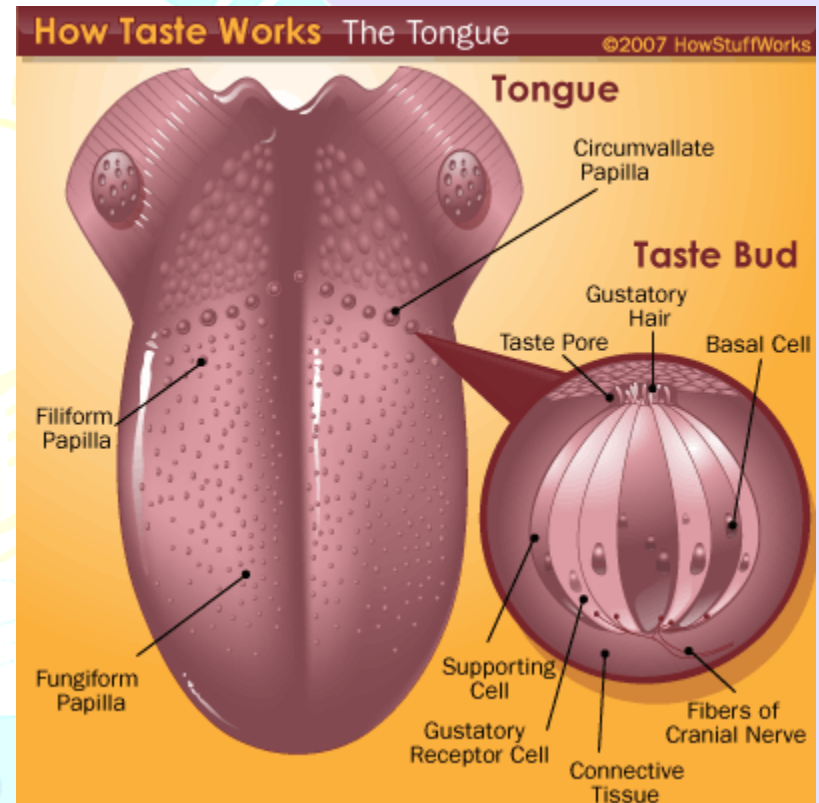
“STRUCTURE OF TASTE BUDS”

© *Each taste bud is flask-like in shape, its broad base resting on the corium, and its neck opening, the gustatory pore, between the cells of the epithelium.*

© *The bud is formed by two kinds of cells:*

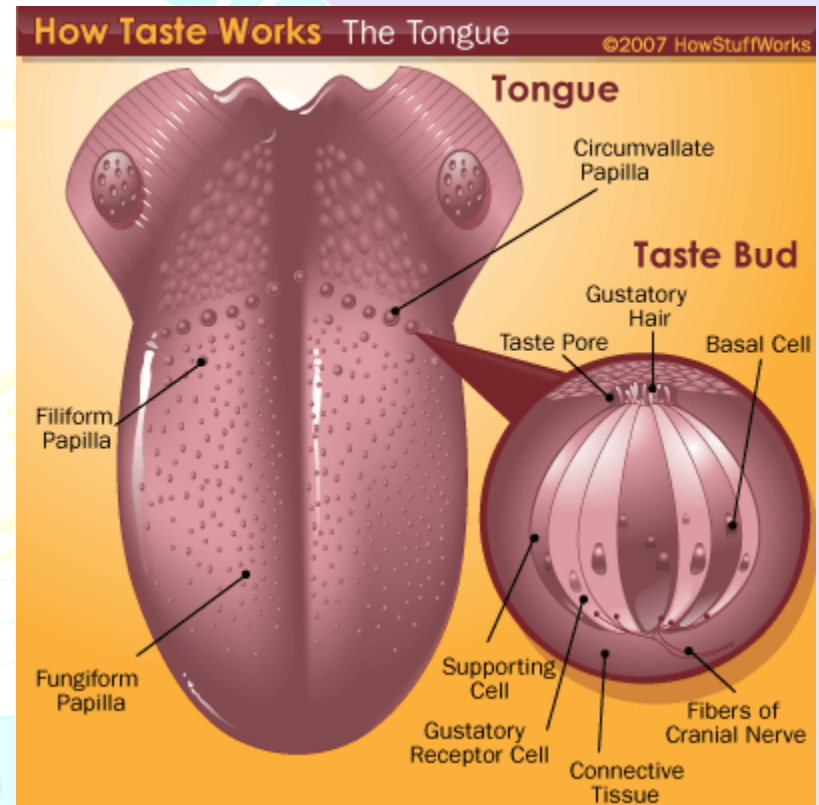
#supporting cells.

#gustatory cells.



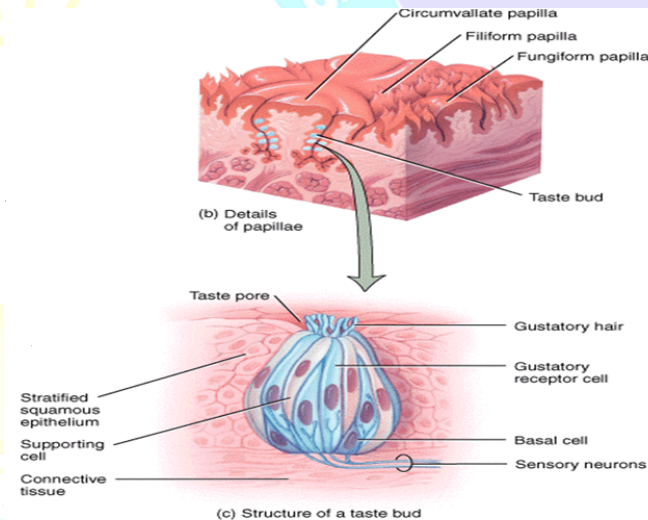
“The supporting (sustentacular) cells”

- ◎ ***Are mostly arranged like the staves of a cask.***
- ◎ ***Form an outer envelope for the bud.***
- ◎ ***Some, however, are found in the interior of the bud between the gustatory cells.***



“The gustatory (taste) cells”

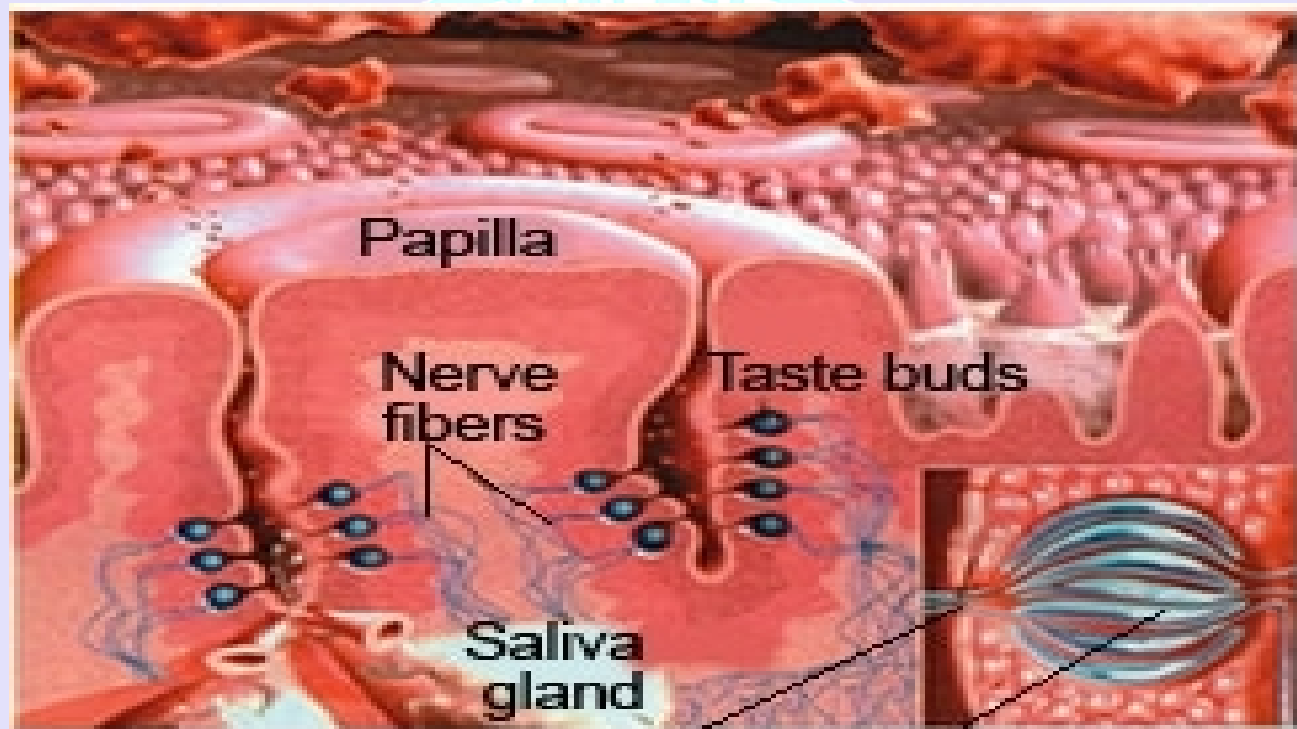
- ◎ ***The gustatory (taste) cells, a chemoreceptor.***
- ◎ ***Occupy the central portion of the bud.***
- ◎ ***They are spindle-shaped.***
- ◎ ***Each possesses a large spherical nucleus near the middle of the cell.***



“STRUCTURE OF TASTE BUDS”

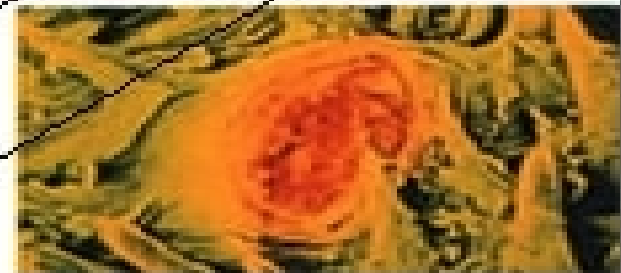
- ◎ *The central process passes toward the deep extremity of the bud, and there ends in single or bifurcated varicosities.*
- ◎ *The nerve fibrils after losing their medullary sheaths enter the taste bud, and end in fine extremities between the gustatory cells.*
- ◎ *other nerve fibrils ramify between the supporting cells and terminate in fine extremities.*
- ◎ *these, however, are believed to be nerves of ordinary sensation and not gustatory.*
- ◎ *The peripheral end of the cell terminates at the gustatory pore in a fine hair filament, the gustatory hair.*

“STRUCTURE OF TASTE BUDS”



Nerve fibers

Taste buds



“TYPES OF PAPILLAE”

◎ ***The majority of taste buds on the tongue sit on raised protrusions of the tongue surface called papillae.***

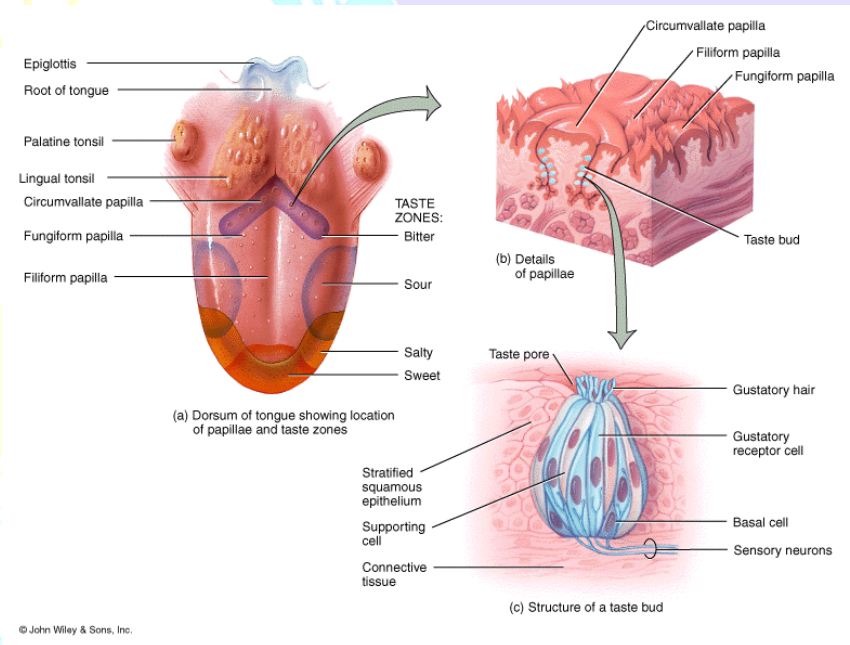
◎ ***There are four types of papillae present in the human tongue.***

Fungiform papillae,

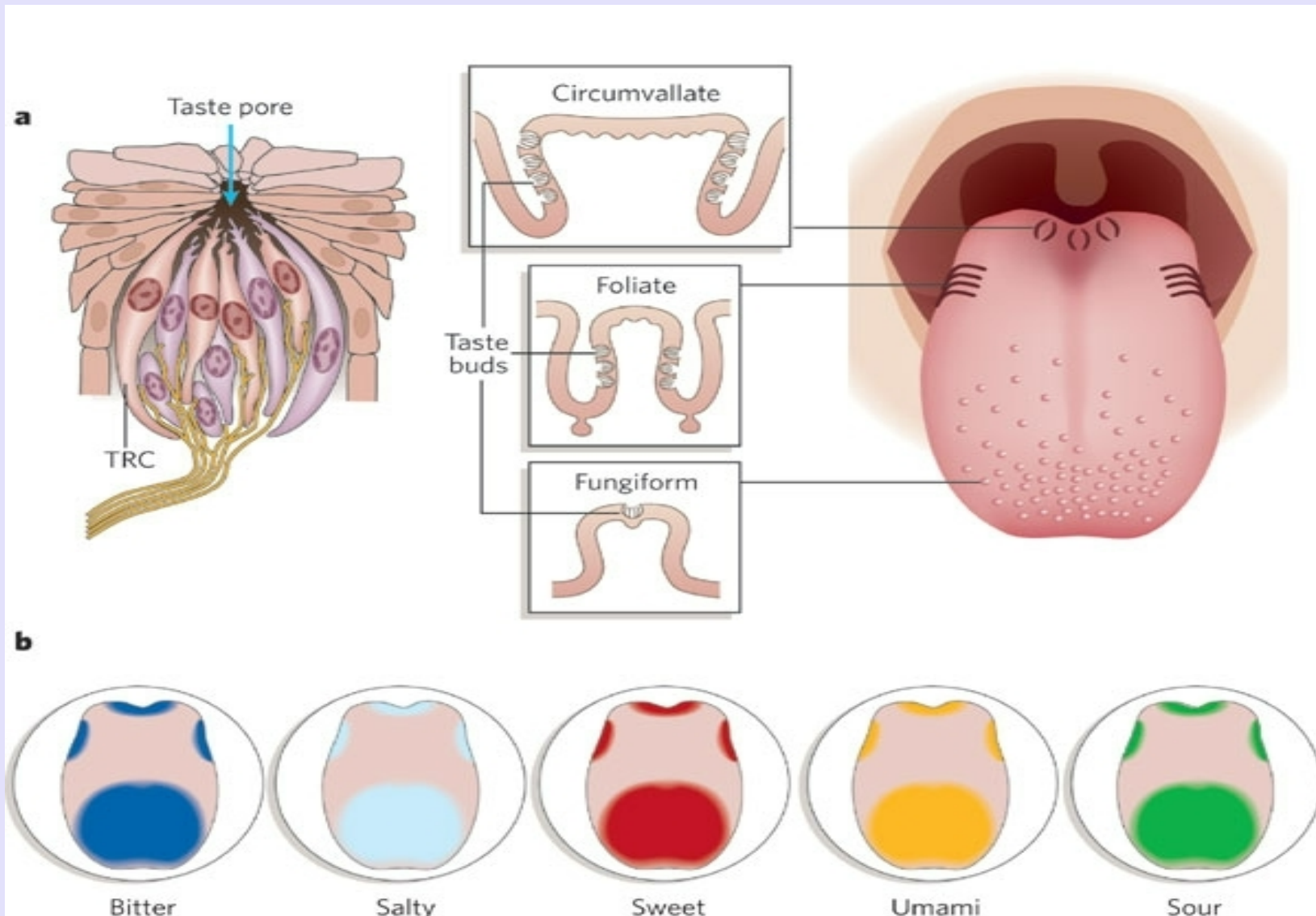
Filiform papillae,

Foliate papillae,

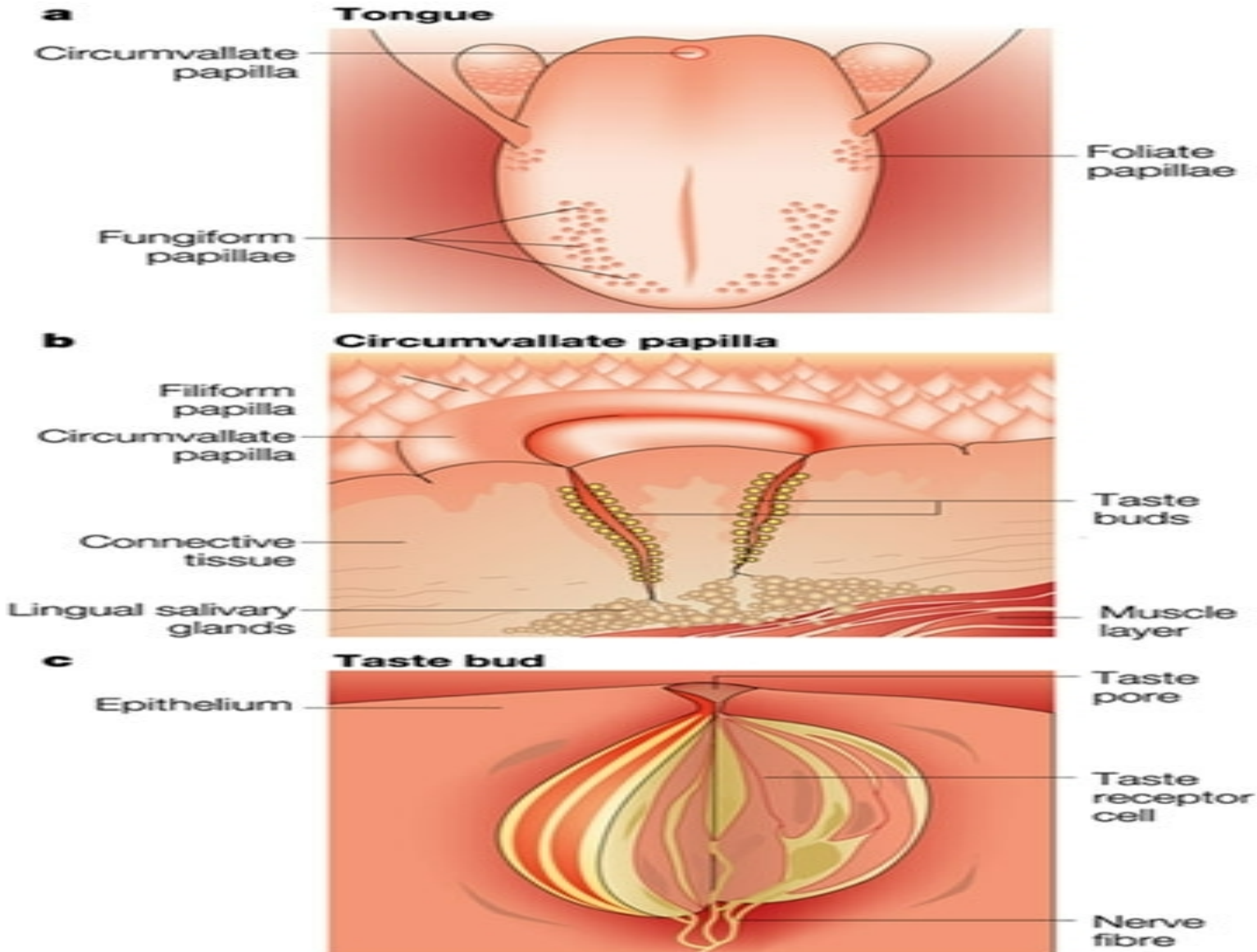
Circumvallate papillae.



“TYPES OF PAPILLAE”

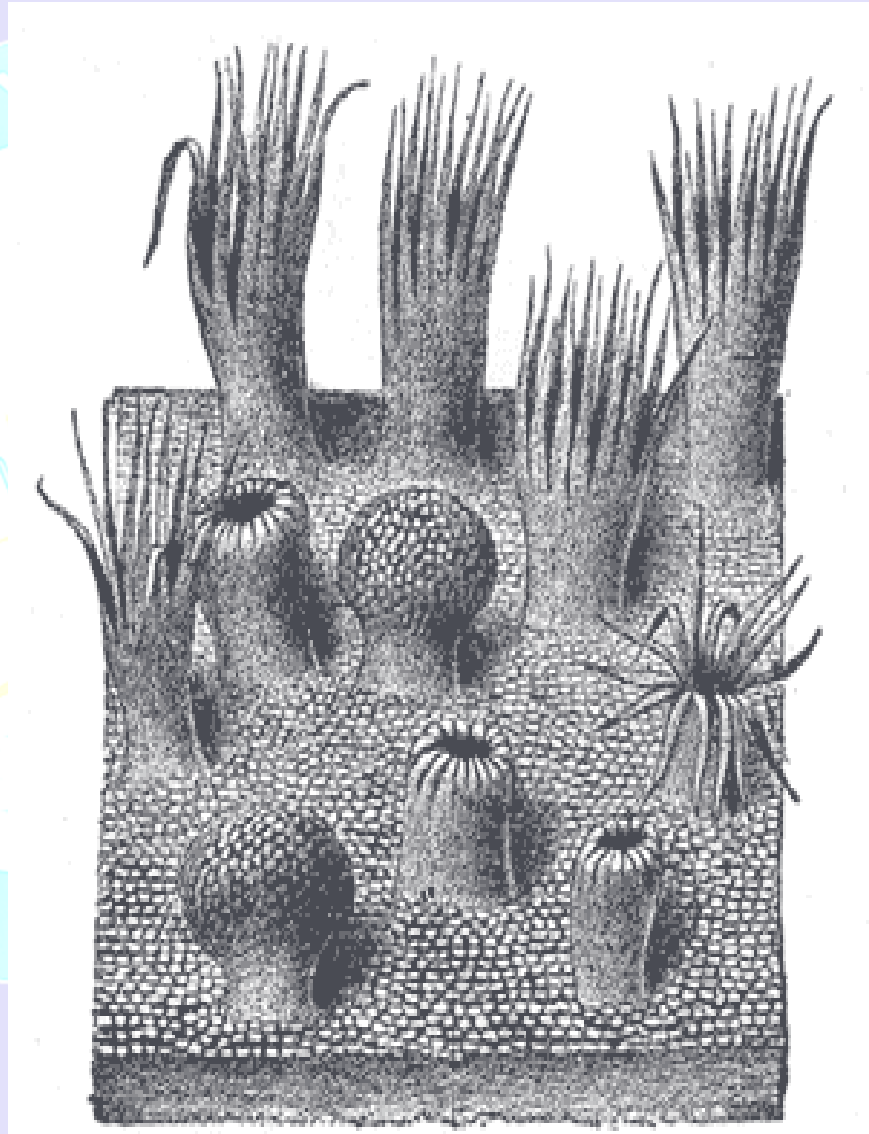


“TYPES OF PAPILLAE”



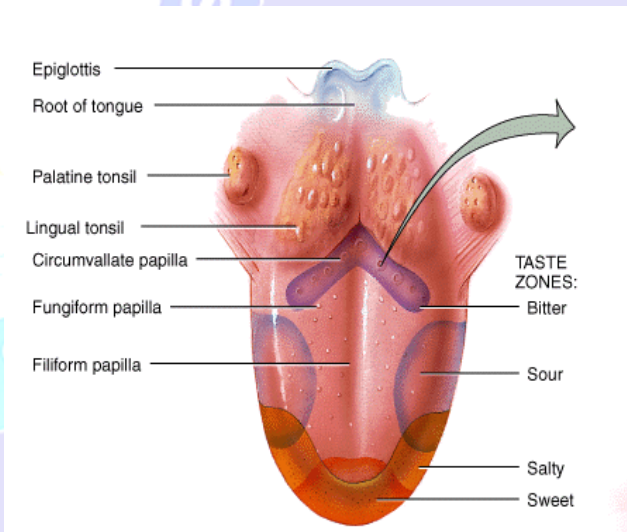
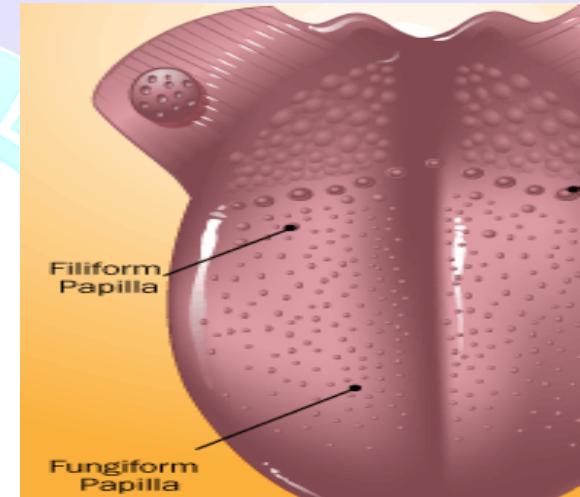
“TYPES OF PAPILLAE”

- ◎ ***View of a portion of the mucous membrane of the tongue.***
- ◎ ***Two fungiform papillæ are shown.***
- ◎ ***On some of the filiform papillae the epithelial prolongations stand erect, in one they are spread out, and in three they are folded in.***



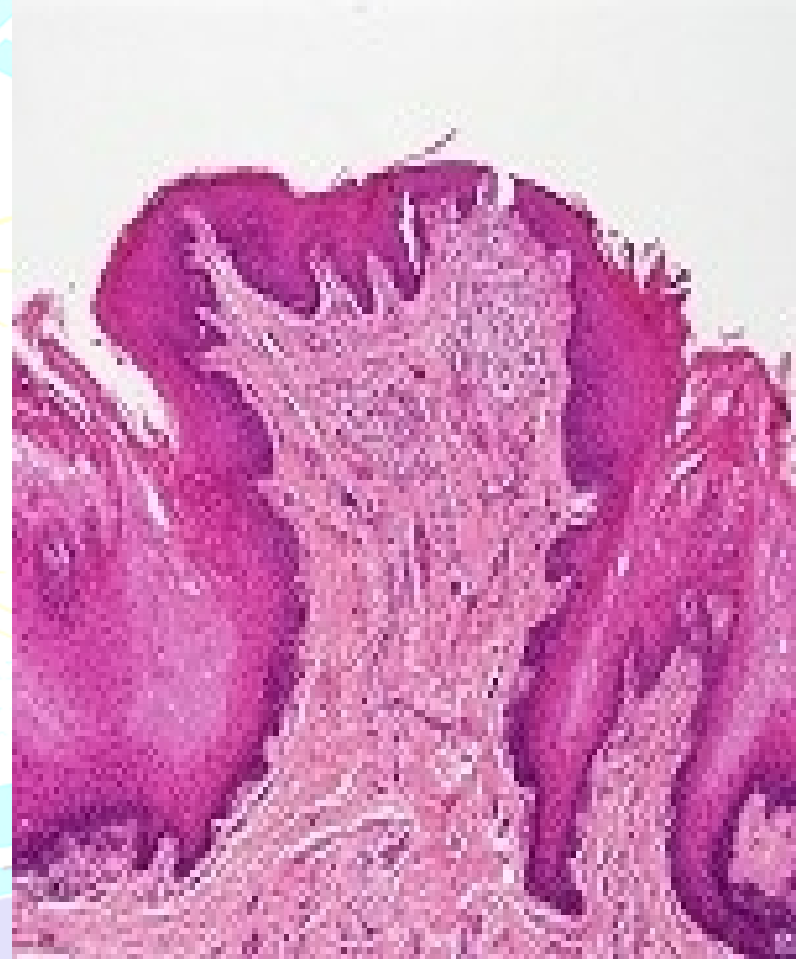
“Fungiform papillae”

- ◎ ***These are slightly mushroom-shaped if looked at in longitudinal section.***
- ◎ ***These are present mostly at the apex (tip) of the tongue, as well as at the sides.***
- ◎ ***Innervated by facial nerve.***



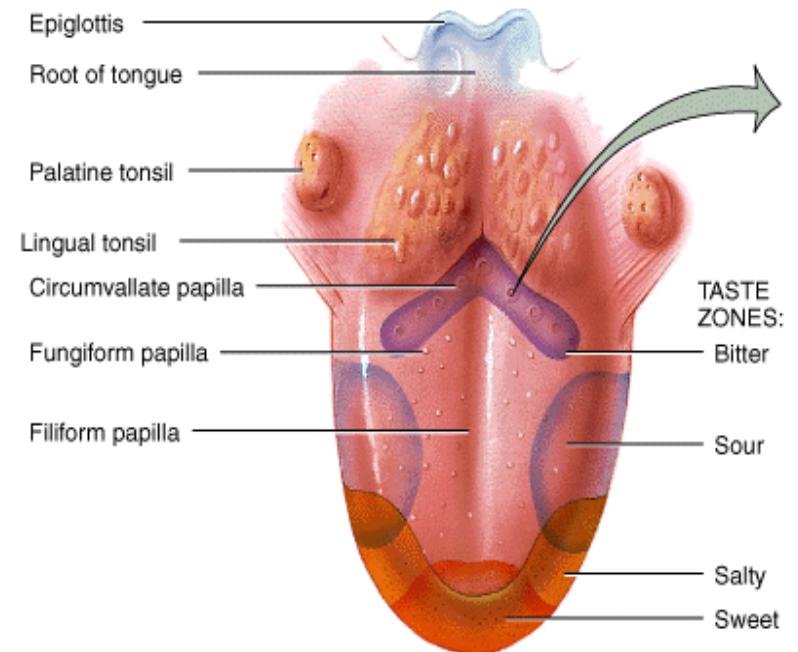
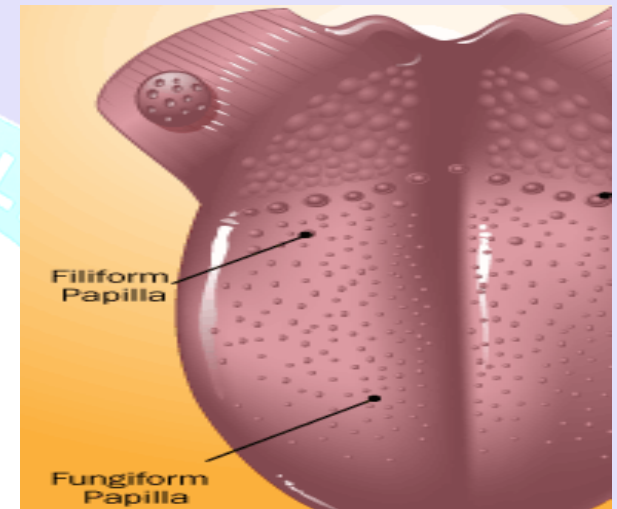
Fungiform Papillae

- *Fungiform papillae occur singly,*
- *They are most numerous on the tip of the tongue.*
- *Central core is made up of vascular connective tissue.*
- *They are covered by stratified squamous non-cornified epithelium.*



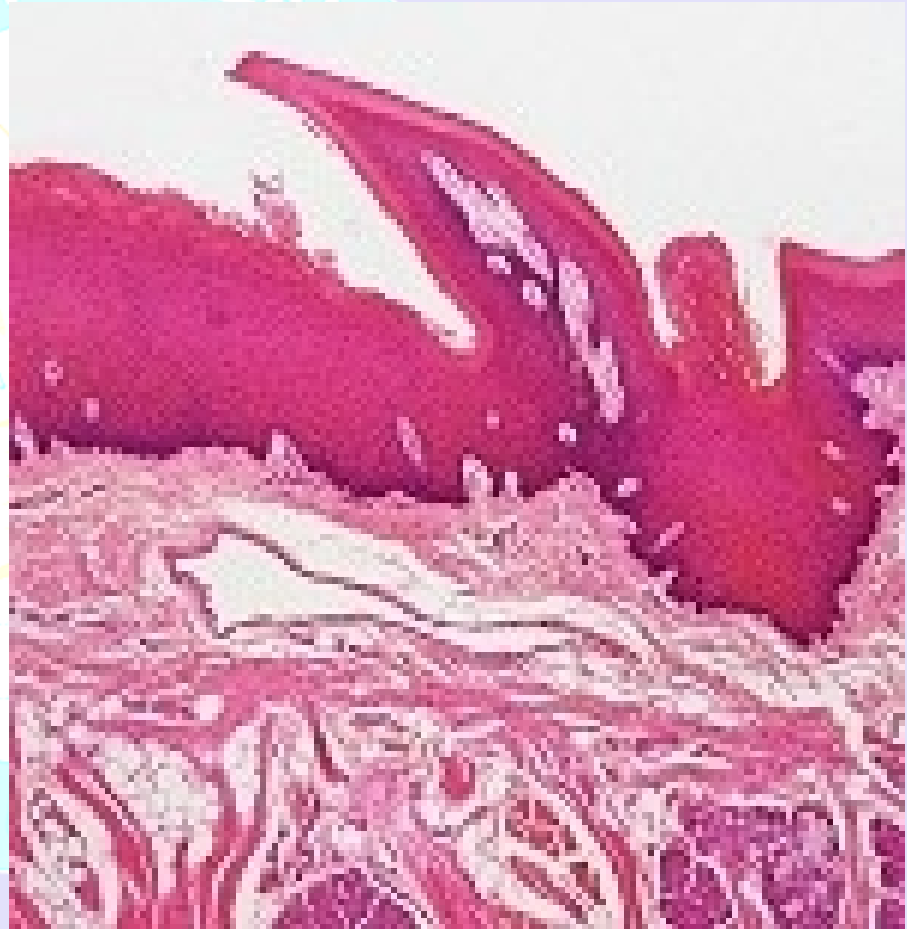
“Filiform papillae”

- *These are thin.*
- *Long papillae .*
- *“V”-shaped cones*
- *That don't contain taste buds but are the most numerous.*
- *These papillae are mechanical and not involved in gustation.*
- *They are characterized by increased keratinization.*



Filiform Papillae

- *Filiform papillae* are the smallest and most numerous papillae.
- They are thread like, covered by stratified squamous cornified epithelium.
- They aid in the manipulation and processing of foods

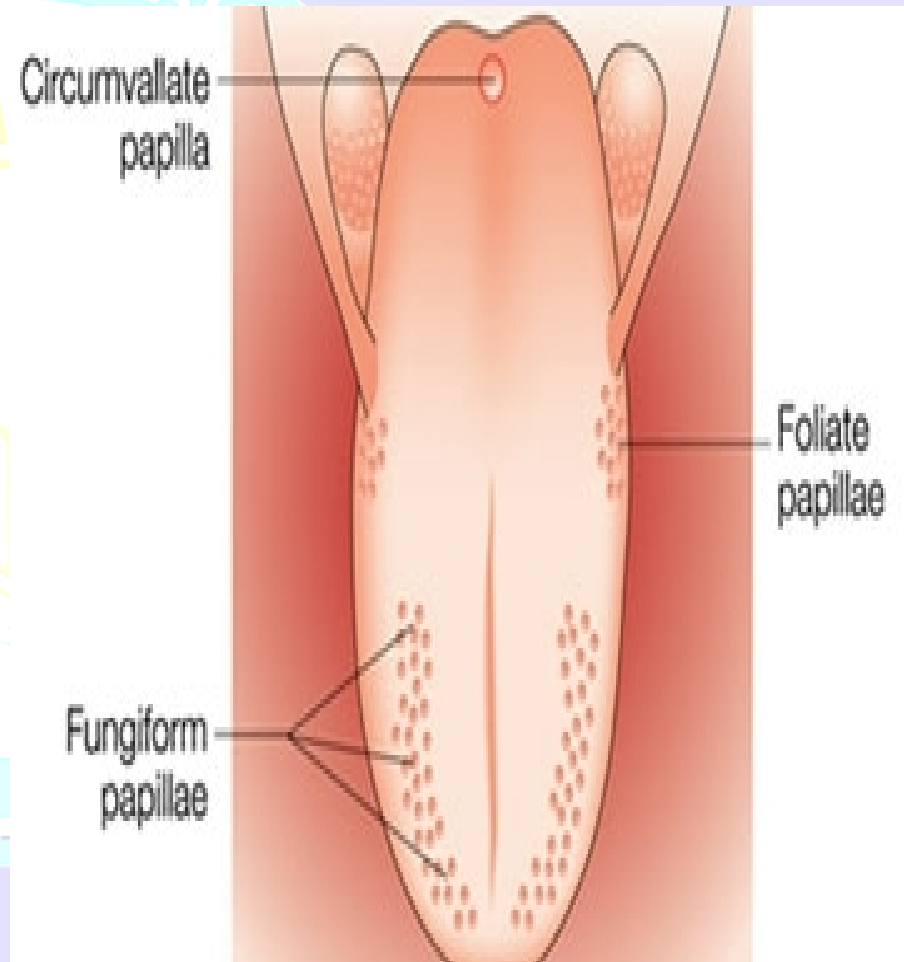


Filiform papillae



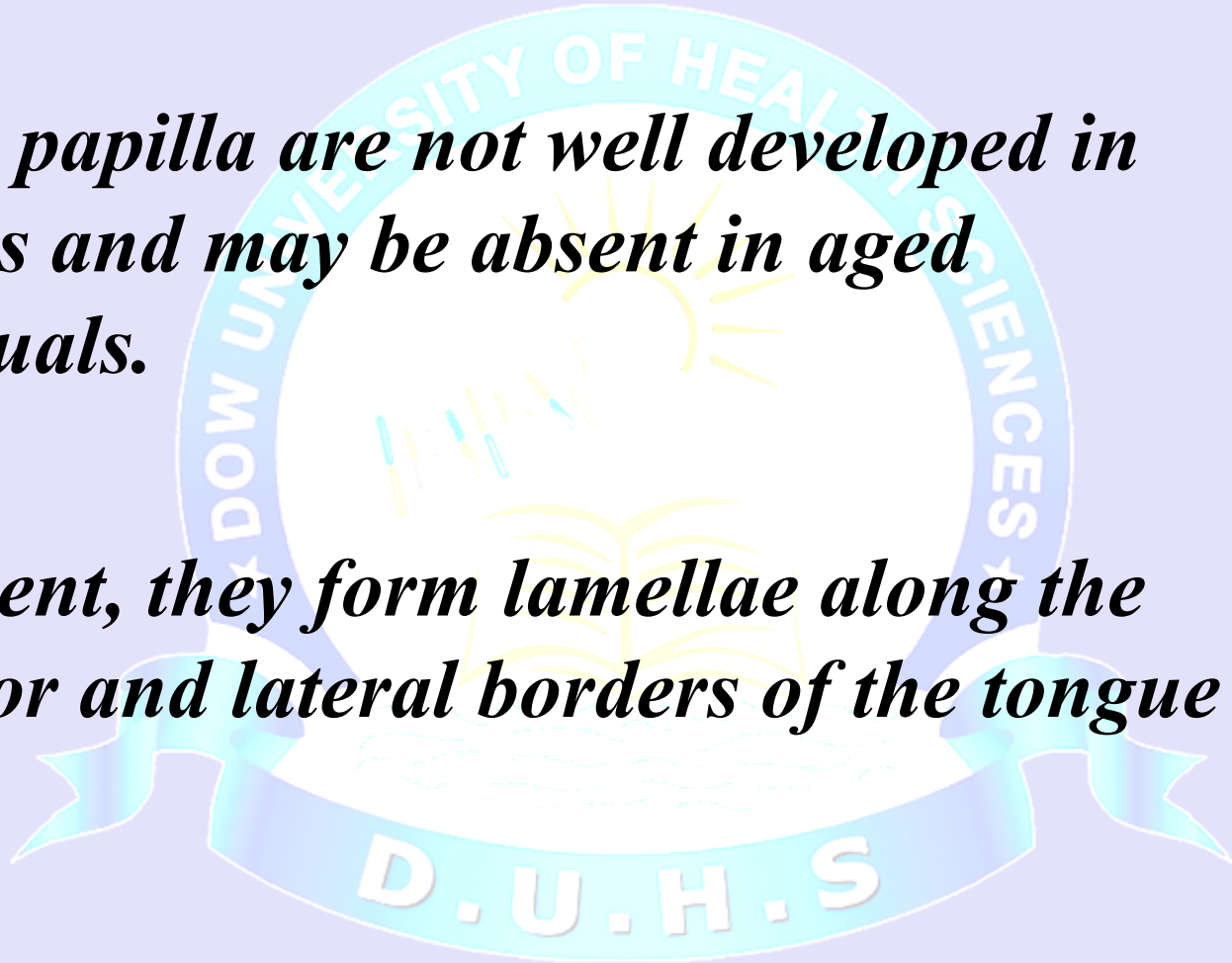
“Foliate papillae”

- ◎ *These are ridges and grooves towards the posterior part of the tongue found on lateral margins.*
- ◎ *Innervated by, facial nerve (anterior papillae) glossopharyngeal nerve (posterior papillae).*



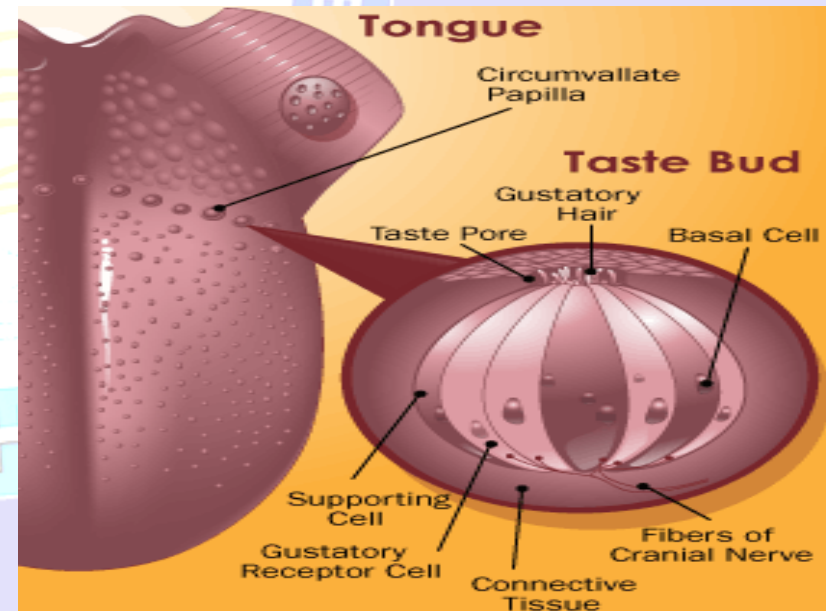
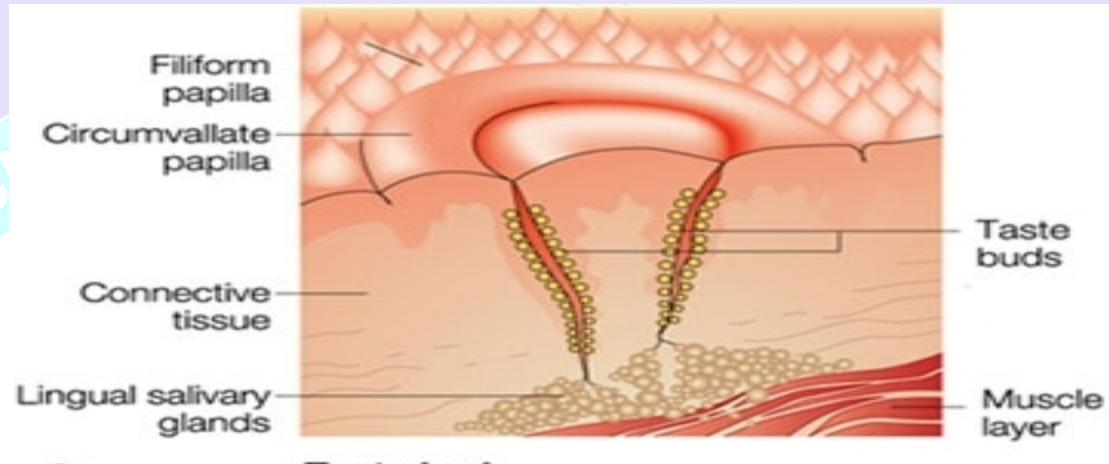
Foliate Papilla

- *Foliate papilla are not well developed in humans and may be absent in aged individuals.*
- *If present, they form lamellae along the posterior and lateral borders of the tongue*



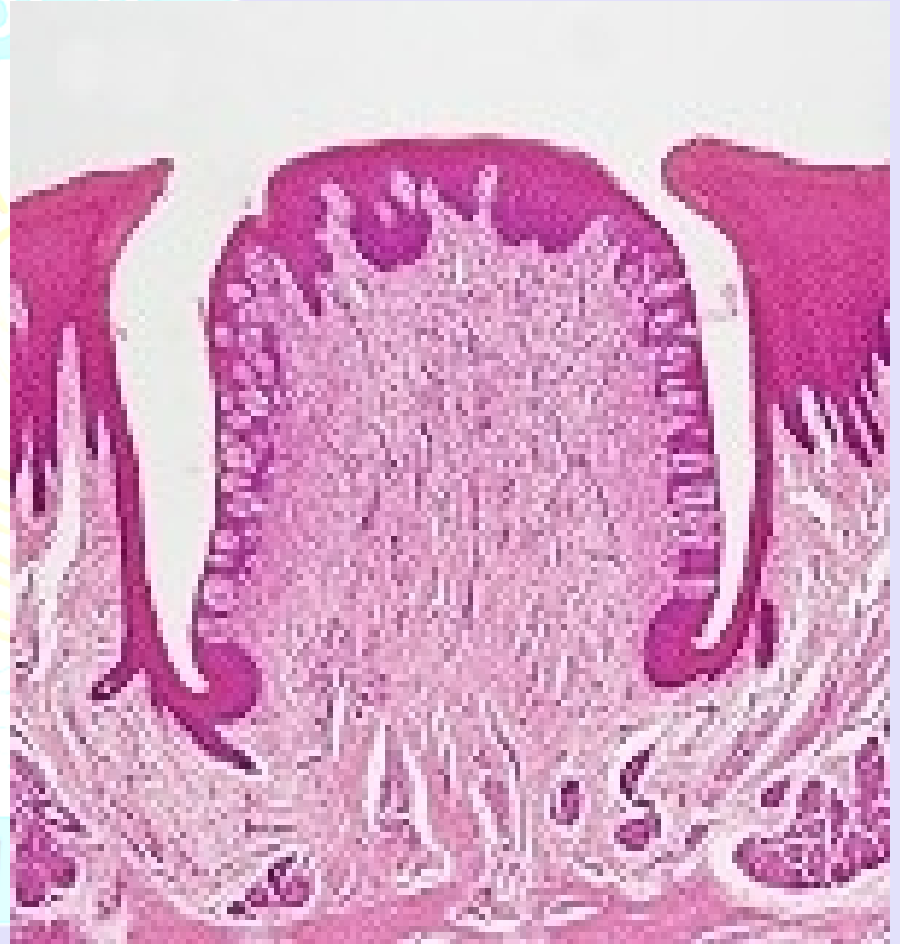
"Circumvallate papillae"

- ⦿ **There are only about 10 to 14 of these papillae on most people.**
- ⦿ **They are present at the back of the oral part of the tongue.**
- ⦿ **They are arranged in a circular-shaped row just in front of the sulcus terminalis of the tongue.**
- ⦿ **They are associated with ducts of Von Ebner's glands.**
- ⦿ **Are innervated by the glossopharyngeal nerve.**



Circumvallate Papillae

- *Circumvallate papillae are the largest and least numerous papillae.*
- *In humans there are between 8 and 12 of them.*
- *They occur in depressions of the surface of the tongue and are surrounded with a trench formed by the infolding of the epithelium.*
- *Taste buds are particularly numerous on the lateral surfaces of these papillae*



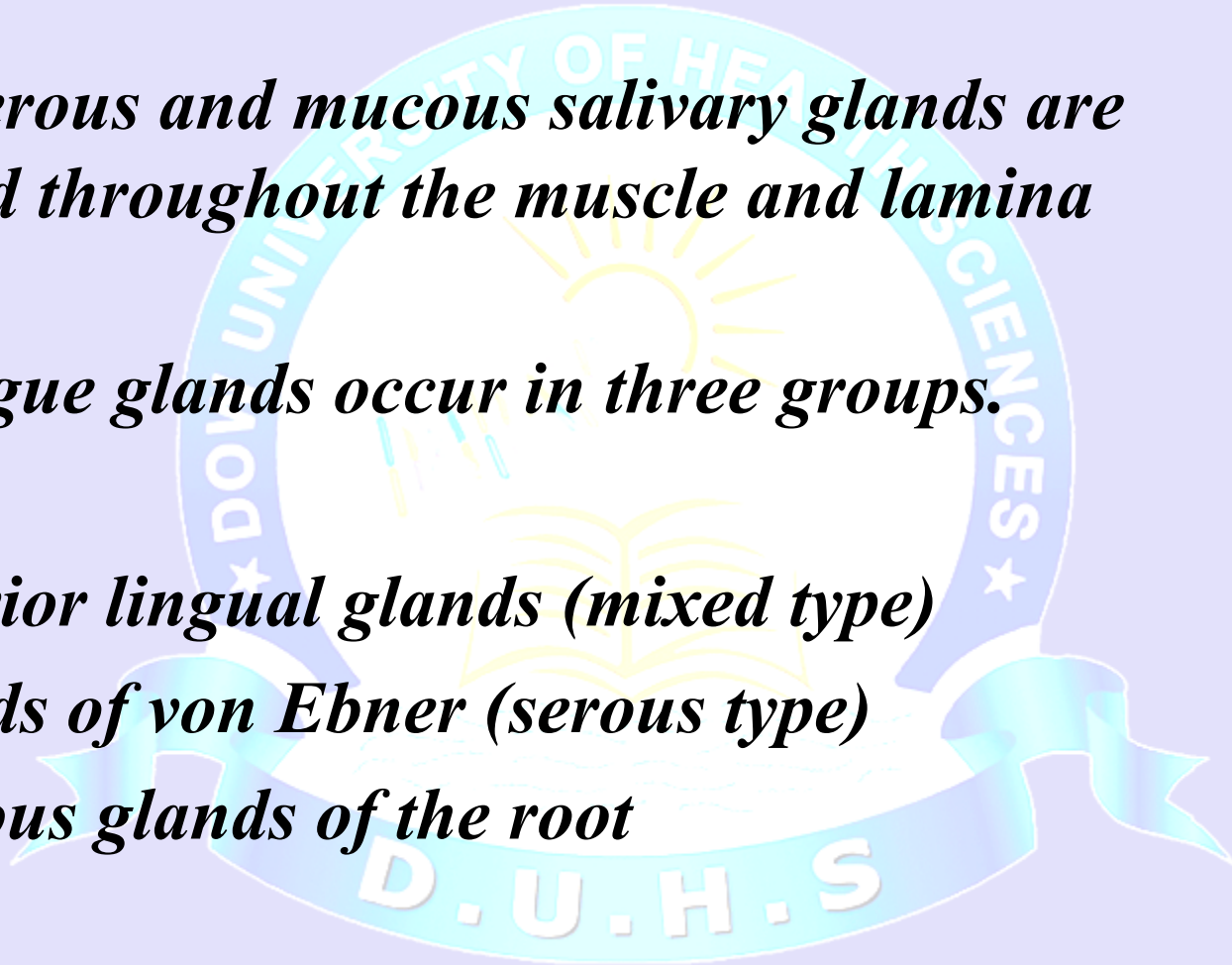
GLANDS OF THE TONGUE

- *Many serous and mucous salivary glands are scattered throughout the muscle and lamina propria.*
- *The tongue glands occur in three groups.*

1- Anterior lingual glands (mixed type)

2- Glands of von Ebner (serous type)

3- Mucous glands of the root



NERVE SUPPLY

- *A number of the cranial nerves supply the tongue.*

Anterior two-thirds

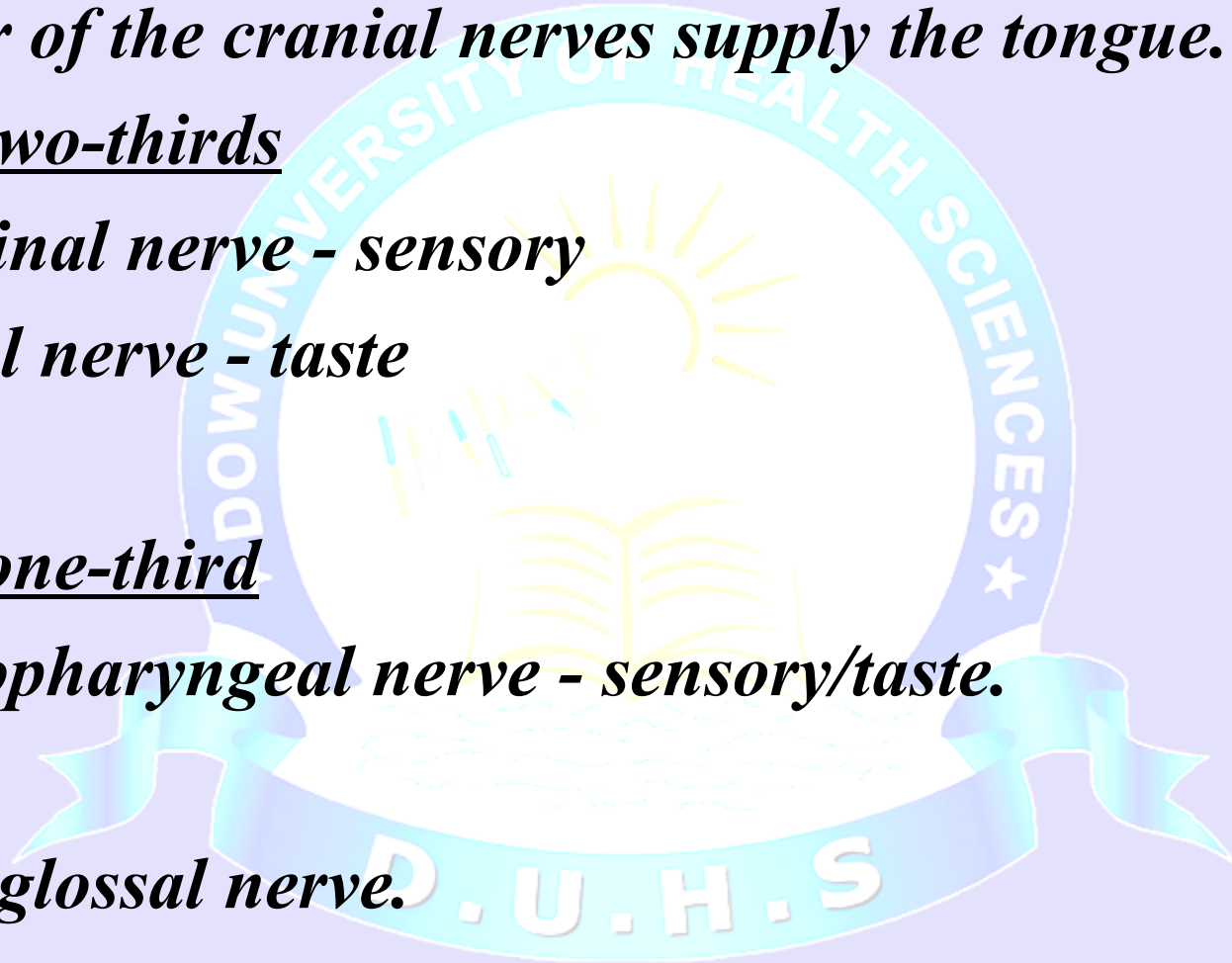
- *V, trigeminal nerve - sensory*
- *VII, facial nerve - taste*

Posterior one-third

- *IX, glossopharyngeal nerve - sensory/taste.*

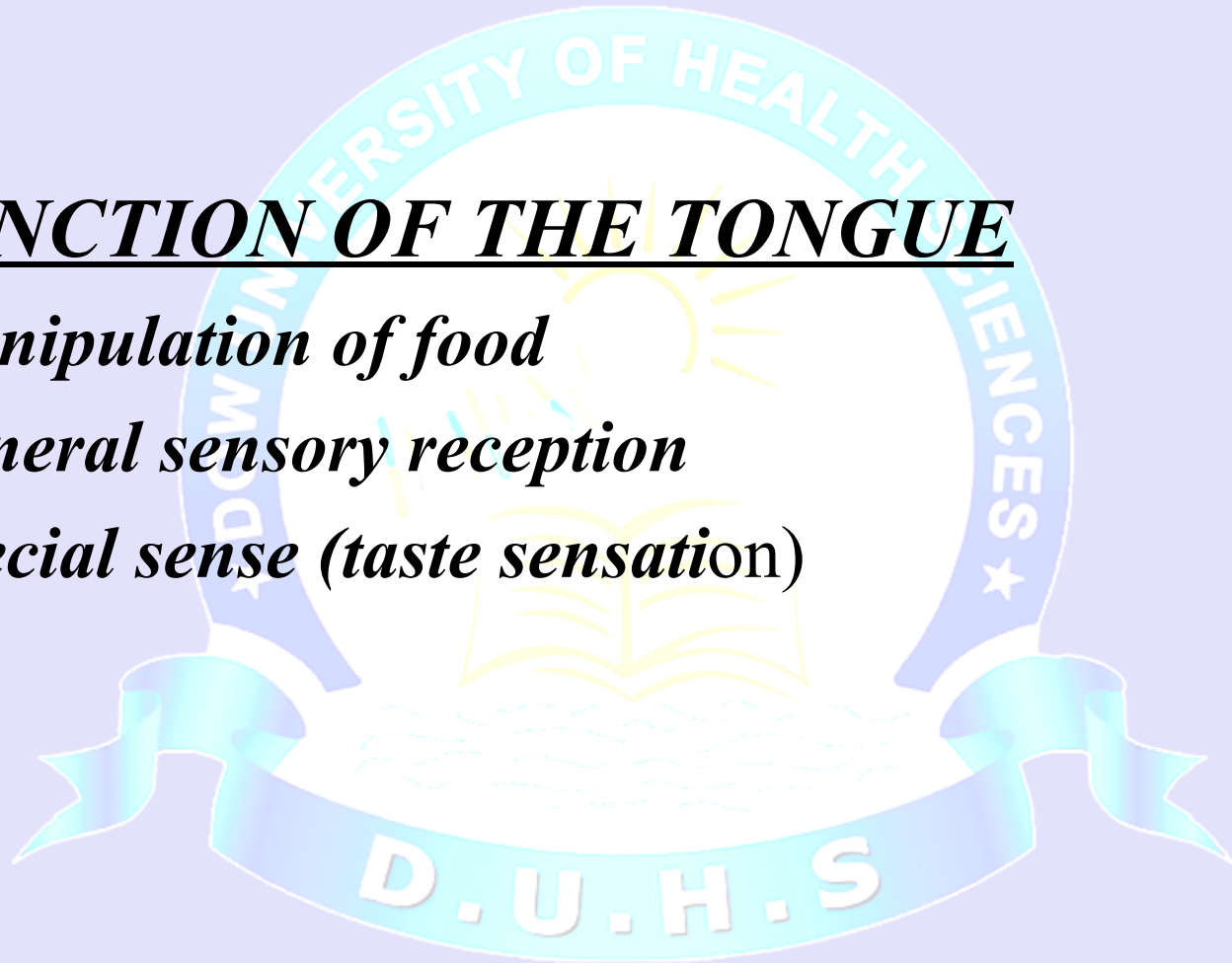
Motor

- *XII, hypoglossal nerve.*



FUNCTION OF THE TONGUE

- *Manipulation of food*
- *General sensory reception*
- *Special sense (taste sensation)*





Thanks